Multiply rational expressions

1 Fully simplify the following expressions:

a $\frac{c}{p} \times \frac{f}{v}$

b $\frac{7x}{6} \times \frac{5y}{2}$

c $\frac{11s}{7t} \times \frac{3r}{5q}$

d $\frac{11}{5y} \times \frac{7}{2y}$

e $\frac{8u}{3v} \times \frac{2v}{7u}$

f $\frac{7x}{4y} \times \frac{7y}{10x}$

g $\frac{x^2}{5} \times \frac{15}{x}$

h $\frac{2n^2}{6} \times \frac{-2y^2}{3n}$

i $\frac{7a^2}{15b^2} \times \frac{10b}{11a}$

j $\frac{5y}{2} \times 6y^5$

k $\frac{m}{3} \times \frac{5m}{9} \times 81q$

l $\frac{n^2}{4} \times \frac{2n^4}{5} \times 20n$

2 Fully simplify the following expressions:

a $\frac{5x+8}{8xy^2} \times \frac{9xy}{25x+40}$

b $\frac{7x+8}{6xy^2} \times \frac{xy}{14x+16}$

c $\frac{p+7}{5} \times \frac{5p-2}{p^2+14p+49}$

d $\frac{8x+72}{x+2} \times \frac{3x^2+6x}{x^2+2x-63}$

e $\frac{4b\,(b+8)}{(b+8)^2} \times \frac{(b+8)\,(b-1)}{6b}$

f $\frac{4m\,(4m+3)}{(4m+3)^2} \times \frac{(4m+3)\,(m-3)}{6m}$

g $\frac{a^2-16}{a\,(a+4)} \times \frac{7a+28}{28\,(a-4)}$

h $\frac{x\,(x-8)}{x^2+16x+64} \times \frac{16\,(x+8)}{-2x+16}$

i $\frac{x^2-4}{36-x^2} \cdot \frac{x^2-8x+12}{8-12x+6x^2-x^3}$

j $\frac{9xy+18x}{x-9} \times \frac{xy-3x-9y+27}{4y^2+8y}$

Divide rational expressions

3 Fully simplify the following expressions:

a $\frac{7x}{6} \div \frac{4}{11y}$

b $\frac{m}{3} \div \frac{m}{2}$

c $\frac{5y}{9} \div \frac{4y}{11}$

d $\frac{33u}{10y} \div \frac{88q}{25y}$

e $\left(-\frac{7}{9}\right) \div \left(-\frac{14n}{27}\right)$

f $\frac{16u}{17v} \div (-14uv)$

g $\frac{u^2}{9} \div \frac{u}{6}$

h $\frac{45x^2yz}{10} \div \frac{9y^2}{5x}$

i $\frac{35t^2u^2}{12v^2w^2} \div \frac{35t^3u}{12w^2y^2}$

j $\frac{77x^5y^3}{9z} \div \frac{55x^8}{y^2z}$

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4 Fully simplify the following expressions:

a $\dfrac{4k-5}{2} \div \dfrac{12k-15}{4}$

b $\dfrac{k-1}{2} \div \dfrac{k^2-1}{6}$

c $\dfrac{k-1}{20} \div \dfrac{k^2-10k+9}{4}$

d $\dfrac{3ab-30a}{7bc+49b} \div \dfrac{7ab-70a}{5bc+35b}$

e $\dfrac{3m^2+3m}{7myz} \div \dfrac{10m+10}{3mz}$

f $\dfrac{6x\,(y+6)}{xy+6x+6y+36} \div \dfrac{36xy}{6x+36}$

g $\dfrac{5x+5}{4\,(x+5)} \div \dfrac{7\,(x-9)}{2x^2-8x-90}$

h $\dfrac{(m+9)\,(m+5)}{20\,(m+8)} \div \dfrac{(m+5)\,(m+7)}{15\,(m+7)}$

i $\dfrac{x-2}{3x^2-30x+63} \div \dfrac{x^2-4}{5x^2-65x+210}$

j $\dfrac{4xz+4yz}{2x-3y} \div \dfrac{24xz-36yz}{4x^2-12xy+9y^2}$

5 Determine whether each of the following is equivalent to $\dfrac{\frac{7}{8}}{x}$:

a $\dfrac{7}{1} \times \dfrac{x}{8}$

b $\dfrac{7}{1} \times \dfrac{8}{x}$

c $\dfrac{7}{1} \div \dfrac{8}{x}$

d $\dfrac{1}{7} \div \dfrac{8}{x}$

6 Determine whether each of the following is equivalent to $\dfrac{\frac{x-5}{8}}{\frac{y+1}{9}}$:

a $\dfrac{x-5}{8} \div \dfrac{y+1}{9}$

b $\dfrac{y+1}{9} \div \dfrac{x-5}{8}$

c $\dfrac{x-5}{8} \times \dfrac{y+1}{9}$

d $\dfrac{x-5}{8} \times \dfrac{9}{y+1}$

7 Complete the following statements:

a $\dfrac{\frac{v}{3}}{\frac{4u}{3}} = \dfrac{\square\left(\frac{v}{3}\right)}{\square\left(\frac{4u}{3}\right)} = \dfrac{\square}{\square}$

b $\dfrac{\frac{u}{4}}{\frac{v}{8}} = \dfrac{\square\left(\frac{u}{4}\right)}{\square\left(\frac{v}{8}\right)} = \dfrac{\square}{\square}$

c $\dfrac{\frac{y}{2}}{\frac{3x}{2}} = \dfrac{\square\left(\frac{y}{2}\right)}{\square\left(\frac{3x}{2}\right)} = \dfrac{\square}{\square}$

d $\dfrac{\frac{4}{n}}{\frac{5}{n^2}} = \dfrac{\square\left(\frac{4}{n}\right)}{\square\left(\frac{5}{n^2}\right)} = \dfrac{\square}{\square}$

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8 Fully simplify the following expressions:

a $\dfrac{\frac{1}{2}}{\frac{3}{4}}$

b $\dfrac{-\frac{9x}{4}}{-\frac{3x}{2}}$

c $\dfrac{\frac{y^2}{6}}{\frac{y}{18}}$

d $-\dfrac{\frac{3}{10x^2}}{\frac{15}{16x^3}}$

e $\dfrac{\frac{4y-6}{10}}{\frac{6y-9}{5}}$

f $\dfrac{\frac{9y-12}{15}}{\frac{12y-16}{5}}$

g $\dfrac{\frac{8x-6}{3x^2}}{\frac{4x-3}{9x}}$

h $\dfrac{\frac{4x-2}{4x^2}}{\frac{2x-1}{8x}}$

Applications

9 The product of two fractions is 1. If one of the fractions is $\dfrac{5x}{9y}$, what is the other fraction?

10 A rectangle has an area of $\dfrac{5x^3y^4}{3pq}$ units^2 and a length of $\dfrac{4xy}{p}$ units. Find an expression for the width of the rectangle.

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